

LAB REPORT : 02

Problem 1

Problem Definition: Simplification of the Boolean expression $F(A,B,C) = A'BC + AB'C + ABC' + ABC$ using Boolean algebra and the Karnaugh map, followed by verification of both the original and simplified circuits using logic simulation.

Objective

To simplify a three-variable sum-of-minterms Boolean expression into its minimal sum-of-products (SOP) form, verify the simplification using a Karnaugh map, and confirm that the original and simplified circuits are functionally equivalent by implementing and testing both circuits in Logisim.

Algebraic Simplification:

$$F(A,B,C) = A'BC + AB'C + ABC' + ABC$$

$$= A'BC + ABC + AB'C + ABC + ABC' + ABC$$

$$= BC(A' + A) + AC(B' + B) + AB(C' + C)$$

$$= BC + AC + AB$$

$$= AB + AC + BC$$

Truth Table:

A	B	C	F
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	1

Table 1: Truth table for $F = AB + AC + BC$

Karnaugh Map:

ABC	00	01	11	10
0	0	0	1	0
1	0	1	1	1

Table 2: K-map for $F(A,B,C) = A'BC + AB'C + ABC' + ABC$

Three groups of two adjacent 1-cells are formed. The vertical pair at $BC = 11$ gives BC . The horizontal pair in the $A = 1$ row for $BC = 01, 11$ gives AC , while the horizontal pair for $BC = 11, 10$ gives AB . Therefore, combining the three groups, the minimized expression is:

$$F = AB + AC + BC$$

This matches the result obtained through algebraic simplification.

Circuit design:

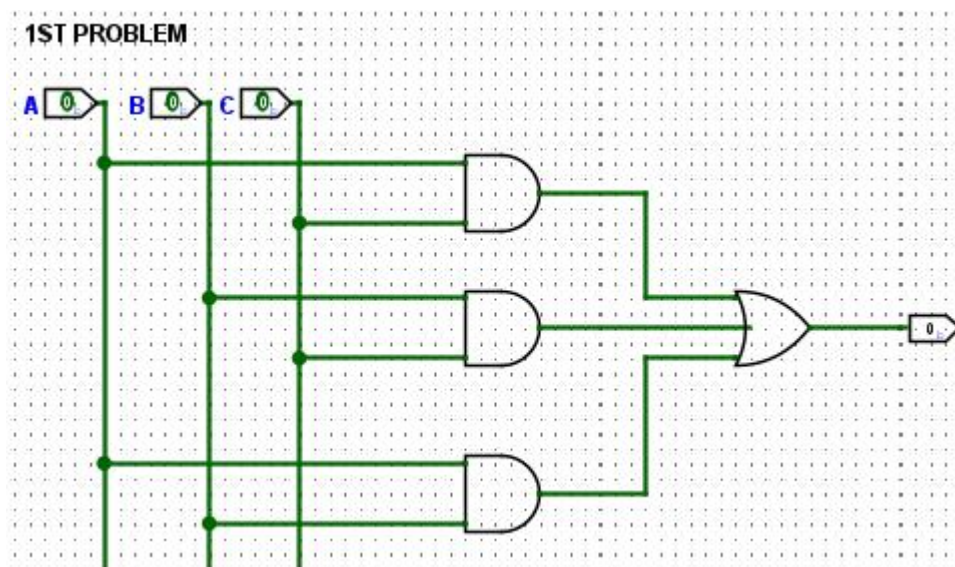


Figure 1: simplified circuit (three 2-input AND gates plus an OR gate, realizing $AB + AC + BC$), both simulated in Logisim

Discussion

The original circuit directly implements the given expression, requiring four 3-input AND gates for the product terms, along with inverters to generate (A') , (B') , and (C') where required. The outputs of these AND gates are then connected to a 3-input OR gate.

The simplified circuit implements $(F = AB + AC + BC)$ using only three 2-input AND gates and one OR gate, eliminating the need for inverters. Both circuits were tested in Logisim using the

same input combinations, and their outputs matched in every case. This confirms that the algebraic and Karnaugh-map simplifications are correct.

Therefore, the simplified circuit is more efficient, as it requires fewer gates, fewer gate inputs, and no inverters.

Conclusion

The Boolean expression ($F = A'BC + AB'C + ABC' + ABC$) was successfully simplified to ($F = AB + AC + BC$) using Boolean algebra and verified using a Karnaugh map. Logisim simulations of both the original and simplified circuits produced identical outputs for all possible input combinations, confirming the correctness of the simplification.

Problem 2

Problem Definition: Simplification of the Boolean expression $F(A,B,C) = A(A + B)(A+B+C)$ using Boolean algebra and the Karnaugh map, followed by verification of both the original and simplified circuits using logic simulation.

Objective

To simplify a three-variable product-of-sums (POS) Boolean expression to its minimal form, verify the simplification using a Karnaugh map, and confirm the functional equivalence of the original and simplified circuits by implementing and testing both in Logisim.

Algebraic Simplification:

$$F(A,B,C) = A(A+B)(A+B+C)$$

$$=(A+AB)(A+B+C)$$

$$=A(A+B+C)+AB(A+B+C)$$

$$=A+AB+AC+AB+AB+ABC$$

$$=A+AB+AC+ABC$$

$$=A(1+B+C+BC)$$

$$=A$$

Truth Table:

A	B	C	F = A
0	0	0	0
0	0	1	0

A	B	C	F = A
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	1

Table 3: Truth table for $F = A$

Karnaugh Map:

All four cells in the $A = 1$ row are 1, while the entire $A = 0$ row is 0. This forms a single group of four adjacent cells covering the whole row, which eliminates both B and C and leaves only $F = A$, exactly matching the algebraic simplification.

A \ BC	00	01	11	10
0	0	0	0	0
1	1	1	1	1

Table 4: K-map for $F(A,B,C) = A(A+B)(A+B+C)$

Circuit design:

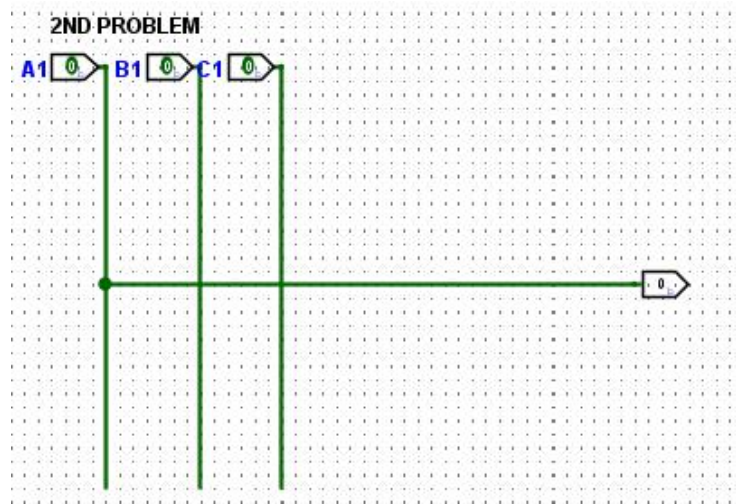


Figure 2: simplified circuit (a direct connection representing $F = A$), simulated in Logisim

Discussion

The original circuit first forms $(A+B)$ using an OR gate, then extends it to $(A+B+C)$ using a second OR gate, and finally ANDs both results with (A) . Since (A) is common to the terms, the expression simplifies to the single literal (A) through the absorption law:

$$A(A+B)(A+B+C)=A$$

Therefore, the simplified circuit consists of only a direct connection from input (A) to the output, requiring no logic gates.

Both the original and simplified circuits were tested in Logisim for all possible combinations of (A) , (B) , and (C) . In every case, the output was equal to (A) , regardless of the values of (B) and (C) . This confirms the correctness of the simplification and demonstrates the application of the absorption law $A(A+B)=A$.

Conclusion

The expression $F = A(A+B)(A+B+C)$ was shown to reduce to $F = A$ through repeated application of the absorption law, and this result was confirmed by both the Karnaugh map and Logisim simulation, which showed that B and C have no effect whatsoever on the output.

Problem 3

Problem Definition: Simplification of the Boolean expression $F(A,B,C) = (A + (BC)')(AB + ABC)$ using Boolean algebra and the Karnaugh map, followed by verification of both the original and simplified circuits using logic simulation an AND gate can be implemented using two NAND gates.

Objective

To reduce a mixed AND-OR-complement expression using De Morgan's theorem and Boolean algebra, to confirm the result with a Karnaugh map, and to verify using Logisim that the simplified circuit behaves as a constant.

Algebraic Simplification:

$$\begin{aligned} F(A,B,C) &= (A+(BC)')(AB + ABC) \\ &= A' \cdot BC \cdot (AB+ABC) \text{ (De Morgan's theorem)} \\ &= A'BC \cdot AB(1+C) \end{aligned}$$

$$=A'BC \cdot AB$$

$$=(AA')BCB$$

$$=0$$

Truth Table:

A	B	C	F
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	0
1	0	0	0
1	0	1	1
1	1	0	0
1	1	1	1

Table 5: Truth table for $F = 0$

Karnaugh Map:

A \ BC	00	01	11	10
0	0	0	0	0
1	0	1	1	0

Table 6: K-map for $F(A,B,C) = (A+(BC)')(AB+ABC)$

Circuit Diagram:

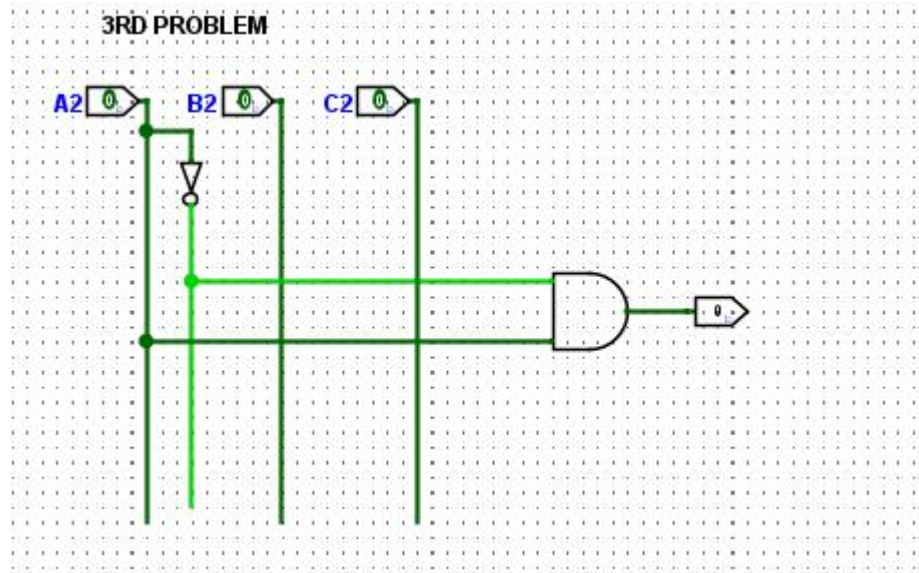


Figure 3: Simplified circuit showing the output permanently at logic 0, simulated in Logisim

Every cell of the map is 0, so there are no 1-cells to group. A Karnaugh map with no populated cells represents the constant function $F = 0$, confirming the algebraic result.

Discussion:

Applying De Morgan's theorem to $(A + (BC)')'$ gives $A'BC$. The second factor simplifies using the absorption law: $AB + ABC = AB(1 + C) = AB$. Therefore, $F = A'BC \cdot AB$. Since the expression contains $A'A = 0$, the entire expression becomes $F = 0$.

Thus, the output is always logic 0, regardless of the values of A, B, and C. This was verified in Logisim by testing all eight possible input combinations. The output remained 0 in every case, which agrees with the all-zero Karnaugh map.

Conclusion:

The expression $F = (A + (BC)')'(AB + ABC)$ was shown to be identically zero for all input combinations. The result was verified algebraically, confirmed by the empty Karnaugh map, and validated through Logisim simulation, which produced a constant logic-0 output for every combination of A, B, and C.

Problem : 04

Problem Definition: Simplification of the Boolean expression $F(A,B) = (B'(A+B) + (A+B)(A+B'))B'$ using Boolean algebra and the Karnaugh map, followed by verification of both the original and simplified circuits using logic simulation.

Objective: To reduce a two-variable expression built from nested OR and AND terms to its minimal form, to confirm the reduction using a Karnaugh map, and to verify the equivalence of the original and simplified circuits in Logisim.

Algebraic Simplification:

$$F(A,B) = (B'(A+B) + (A+B)(A+B'))B'$$

$$= (AB' + BB' + AA + AB' + AB' + BB')B'$$

$$= (A + AB' + AB' + AB')B'$$

$$= A(1 + B' + B' + B')B'$$

$$= AB'$$

Truth Table:

A	B	F = AB'
0	0	0
0	1	0
1	0	1
1	1	0

Table 7: Truth table for $F = AB'$

Karnaugh Map:

A \ B	0	1
0	0	0
1	1	0

Table 8: K-map for $F(A,B) = (B'(A+B) + (A+B)(A+B'))B'$

Only a single cell, at $A = 1, B = 0$, is populated. A lone 1-cell cannot be combined with any adjacent cell, so it is read directly as the product of its coordinate literals, giving $F = AB'$, matching the algebraic result.

Circuit Diagram:

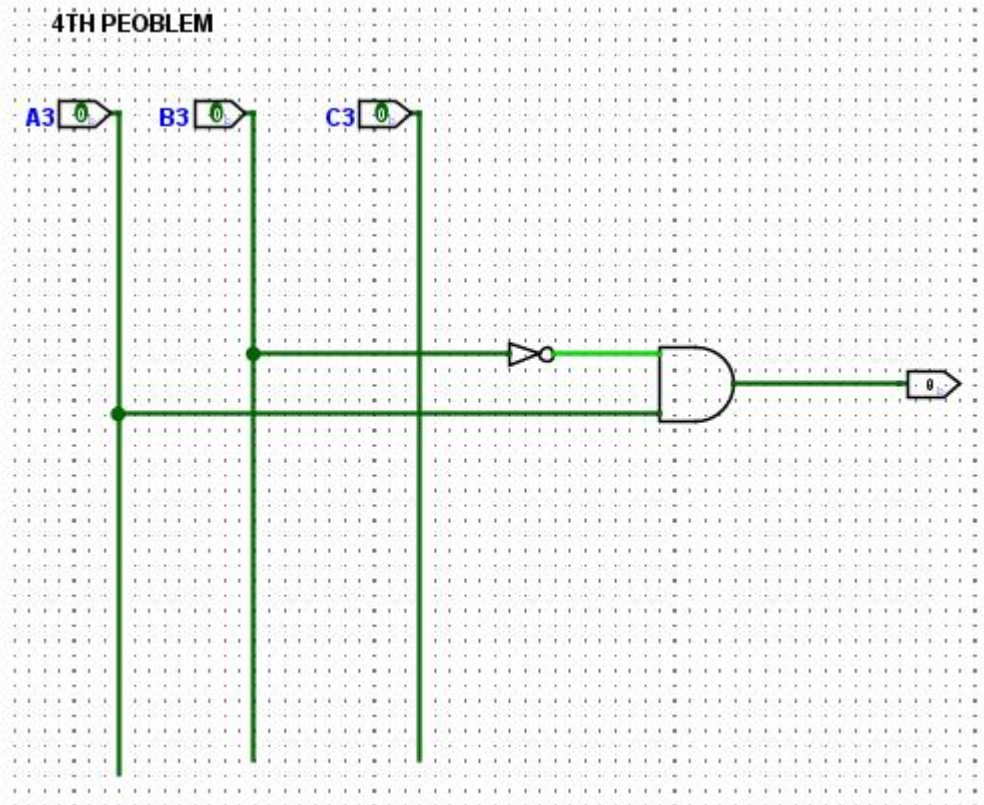


Figure 4:d simplified circuit reduced to a single AND gate realizing AB' , simulated in Logisim

Discussion:

Expanding the original expression term by term produces several instances of AB' along with the terms A , BB' , and AA . Since $BB' = 0$ and $AA = A$, the expression simplifies to AB' . Therefore, the original four-gate network, consisting of OR, AND, and NOT gates as shown in the Logisim schematic, is functionally equivalent to a single 2-input AND gate with B inverted.

Both circuits were tested in Logisim for all four possible combinations of A and B . The output was 1 only when $A = 1$ and $B = 0$, matching $F = AB'$ in every case. This confirms the functional equivalence of the original and simplified circuits.

Conclusion:

The Boolean expression $F = (B'(A + B) + (A + B)(A + B'))B'$ was successfully simplified to $F = AB'$ using Boolean algebra. The result was confirmed using a Karnaugh map and validated through Logisim simulation. Both the original and simplified circuits produced identical outputs for all possible input combinations of A and B.

References:

1. M. M. Mano, Digital Design, 5th ed. Upper Saddle River, N.J.: Pearson, 2013.
2. R. J. Tocci, N. S. Widmer, and G. L. Moss, Digital Systems: Principles and Applications, 11th ed. Upper Saddle River, N.J.: Pearson, 2011.